LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 5: Jump!

Module 2 | 45 minutes | Jumping with velocity | Grades 3–5 · ages 9–11

Objective: Make a sprite jump by setting its velocity upward, only when it is standing on the floor.

Success criteria

- I can make a sprite jump by setting its velocity.
- I can stop mid-air jumps.
- I can tune jump height.

Key vocabulary

- **impulse**: A single push that changes velocity.
- **grounded**: Standing on the floor.
- **arc**: The curved path of a jump.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Ball toss (5 min): toss a soft ball up and describe its speed at the start, top and end.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Make the frog jump: learners follow the steps.
14 min	**Perform**	Code Lab: jump, Debug the flying frog, Anatomy of a jump
9 min	**Probe**	Code Lab: run and jump
4 min	**Wrap-up**	Exit ticket: Which block makes the jump start?

Unplugged warm-up

Ball toss (5 min): toss a soft ball up and describe its speed at the start, top and end.

Teaching the idea

A jump is an impulse: setting the vertical velocity positive once. The existing gravity loop produces the parabolic arc. Grounded checks prevent infinite mid-air jumps.

Analogy: Throwing a ball straight up: one push, then gravity slows it, stops it and brings it back.

Live demo: Jump without the grounded check; press space repeatedly to fly away. Then add the check.

Questions to ask

Ask	Listen for
What is vy at the top of a jump?	0.
How do you jump higher?	Set vy to a bigger number.
Why check the floor before jumping?	To stop mid-air jumps.

Differentiation

Support
Give the jump script with the if; learners choose the numbers.

Stretch
Double jump: allow one extra jump in the air using a jumps-left variable.

Common misconceptions

Misconception	How to fix it
A jump needs its own loop to move up	Set vy once; the gravity loop moves it.
Change y by 14 is a jump	That teleports; velocity makes a smooth arc.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which block makes the jump start?

Homework
Watch a platform game. Does the character jump higher if you hold the button? How might that be coded?

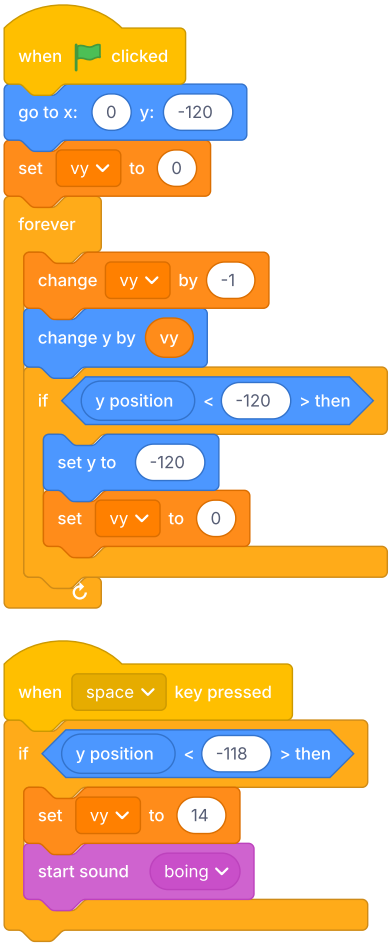
Answer key — Kitabash activities

1. Code Lab: jump (perform · code lab)

Goals checked: Space makes it jump · Lands again · Only jumps from the floor

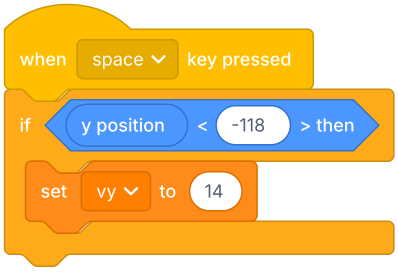
One solution:

Hero



2. Debug the flying frog (perform · multiple choice)

Answer: A



3. Anatomy of a jump (perform · ordering)

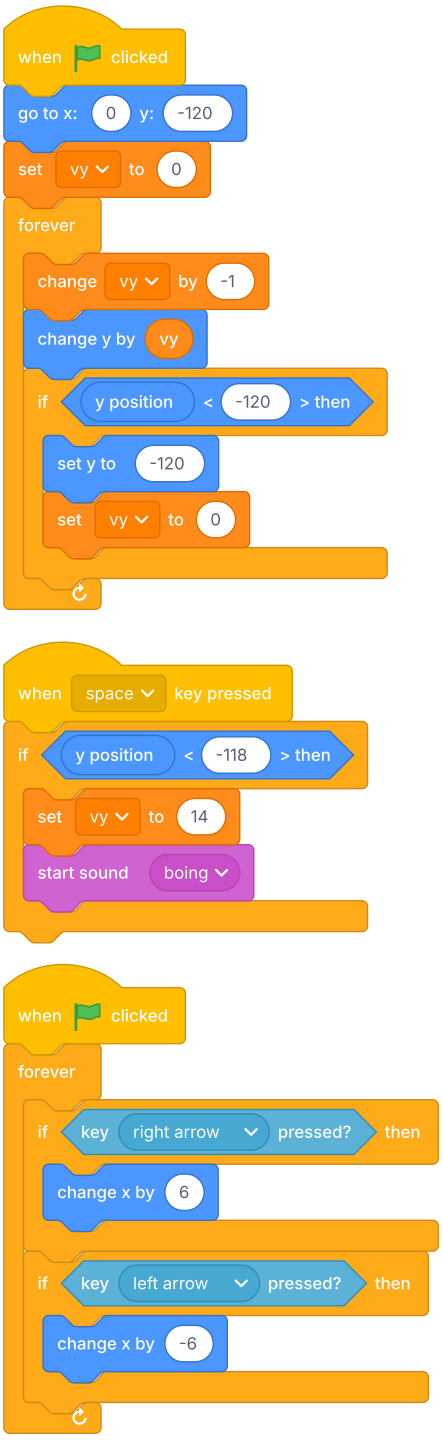
Order: Space pressed: vy is set to 14 → The sprite rises; gravity makes vy smaller each frame → vy reaches 0 at the top of the jump → vy goes negative and the sprite falls → The sprite lands and vy is set to 0

4. Code Lab: run and jump (probe · code lab)

Goals checked: Walks right · Still jumps

One solution:

Hero



Answer key — worksheet

1. **Key words:** impulse = A single push that changes velocity. · grounded = Standing on the floor. · arc = The curved path of a jump.
2. **Plan your code:** Goals: Space makes it jump; Lands again; Only jumps from the floor.
3. **Debug the flying frog:** A — Only allow a jump when the sprite is on the floor.
4. **Anatomy of a jump:** 1 Space pressed: vy is set to 14 · 2 The sprite rises; gravity makes vy smaller each frame · 3 vy reaches 0 at the top of the jump · 4 vy goes negative and the sprite falls · 5 The sprite lands and vy is set to 0
5. **Plan your code:** Goals: Walks right; Still jumps.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.